Lecture 03

Coasean Bargaining

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AEM 4510

Roadmap

1. Can we achieve the efficient outcome **without** government intervention?
2. What does the Coase theorem say?
3. What are the limits to Coasean bargaining?
4. Can data centers and their neighbors bargain their way out of the AI boom's externalities?

This lecture's running example: the AI boom next door

Data centers are a textbook externality problem

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Today's question: when can the operator and the neighbors bargain to the efficient outcome, and when do transactions costs get in the way?

Coase Theorem

Pigou vs Coase

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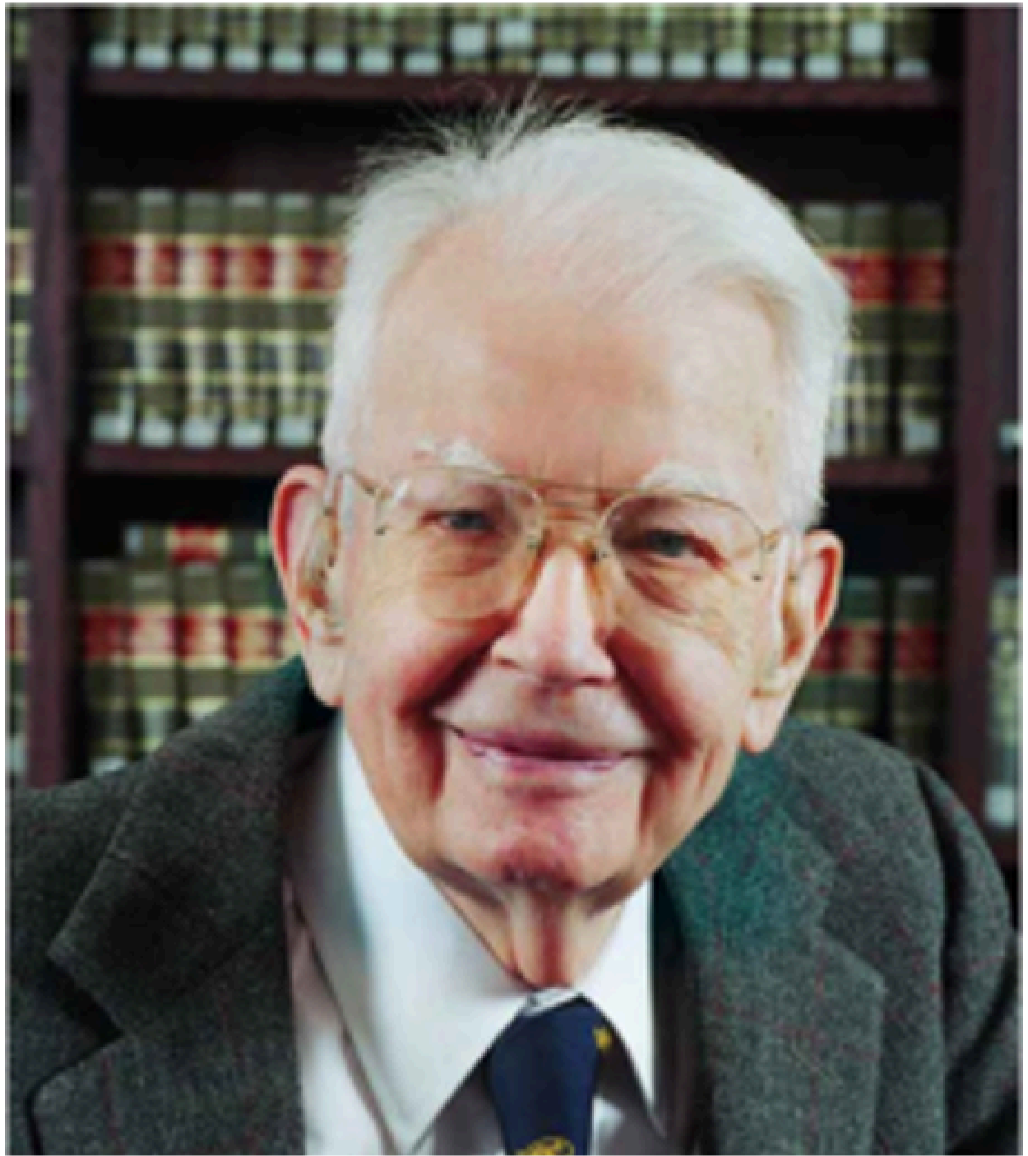
Why?

There are no markets through which the source of the externality must pay/be compensated for its effect on society

i.e. they're not priced

This means there's a role for government to create this market or price the externality

Ronald Coase (1910-2013)



In a famous paper ("The Problem of Social Cost"), 1991 Nobel prize winner Ronald Coase made people rethink this

Do we actually **NEED** government intervention?

The Coase Theorem

If there are:

1. No wealth effects on demand
2. No transactions costs
3. Well-defined and enforceable property rights

then:

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The right to pollute (a resource) will end up in the hands who value it most through negotiation

Property rights

Property rights are only as good as prevailing norms or enforcement

Coasean arguments

Coase versus Pigou: externalities are reciprocal in nature

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A power plant produces emissions that nearby residents breathe and those people incur the external costs

Coasean arguments

Coase versus Pigou: externalities are reciprocal in nature

A power plant produces emissions that nearby residents breathe and those people incur the external costs

By breathing the air, the nearby residents help create the externality (i.e. if they weren't there, there would be no external cost from emitting pollution)

Coasean arguments

What is more valuable to society?

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The externality-producing good generated by the power plant or letting people live nearby?

To make the reciprocity concrete, consider a doctor working next door to a data center whose equipment creates noise

The Doctor and the Data Center

More noise = more computing services and less medical services

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More noise = more computing services and less medical services

Less noise = fewer computing services and more medical services

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Which is better from a social point of view depends upon the relative values of computing and medical services

The Doctor and the Data Center

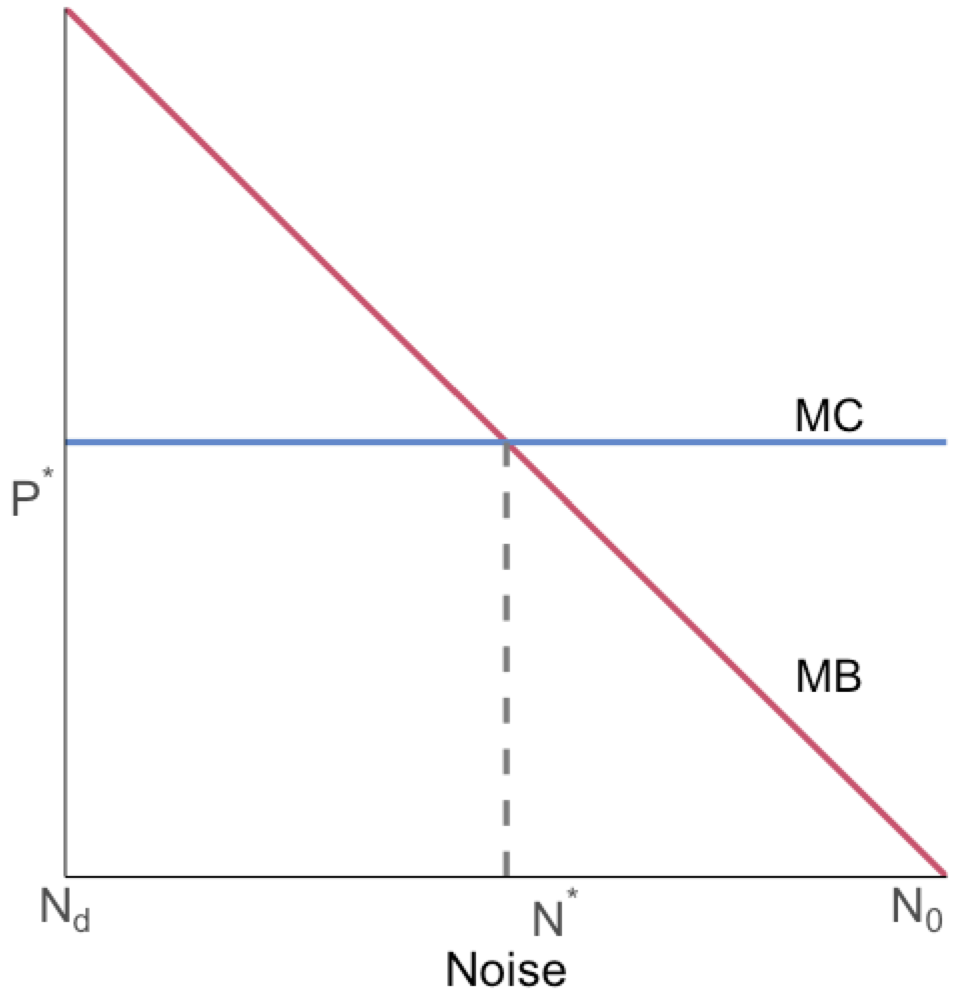
More noise = more computing services and less medical services

Less noise = fewer computing services and more medical services

Which is better from a social point of view depends upon the relative values of computing and medical services

Is the net benefit to society better at no noise, 0, or the level of noise that maximizes the data center's profit, N_0?

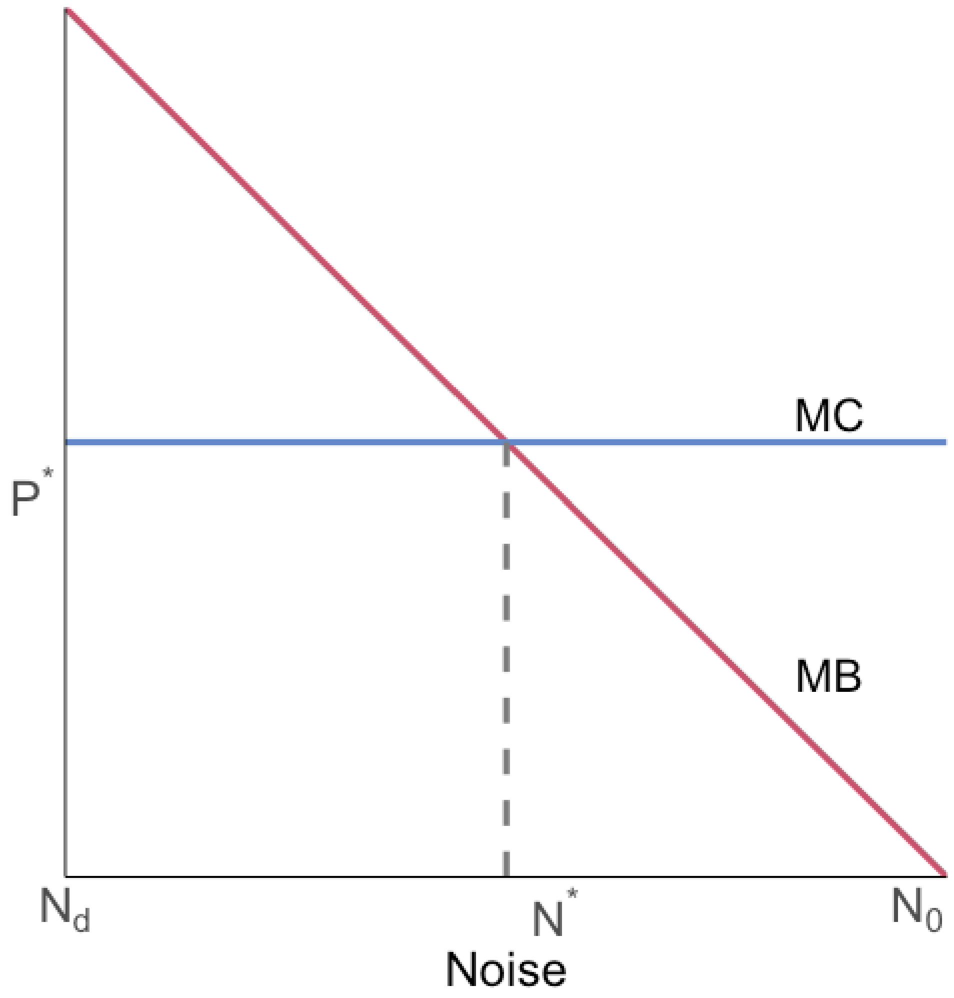
The Doctor and the Data Center



MC is the marginal cost imposed on the doctor by noise

MB is the marginal benefit to the data-center operator from running the equipment that creates noise

Coase: Point 1



It is important to establish that someone has the property rights

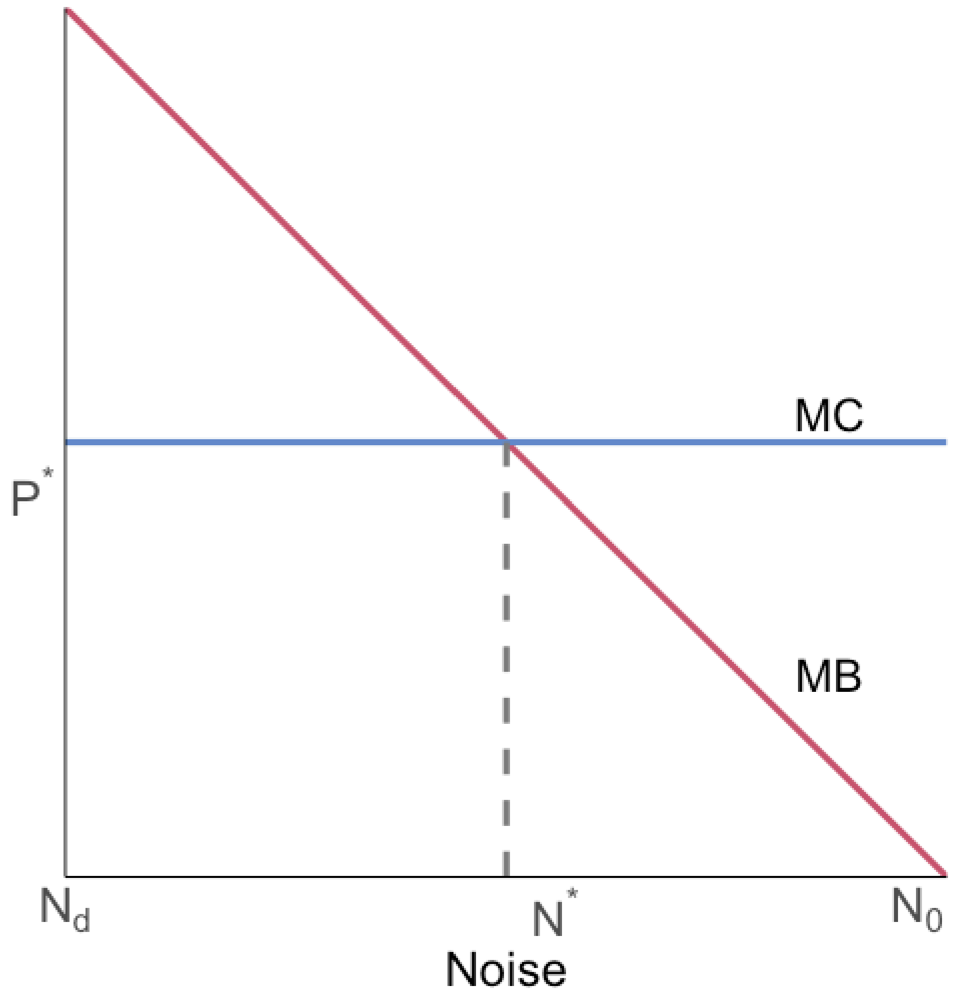
Otherwise, trade will not happen

Give the right to create noise to the data-center operator

Initial outcome will be $N=N_0$

What happens next?

Coase: Point 1

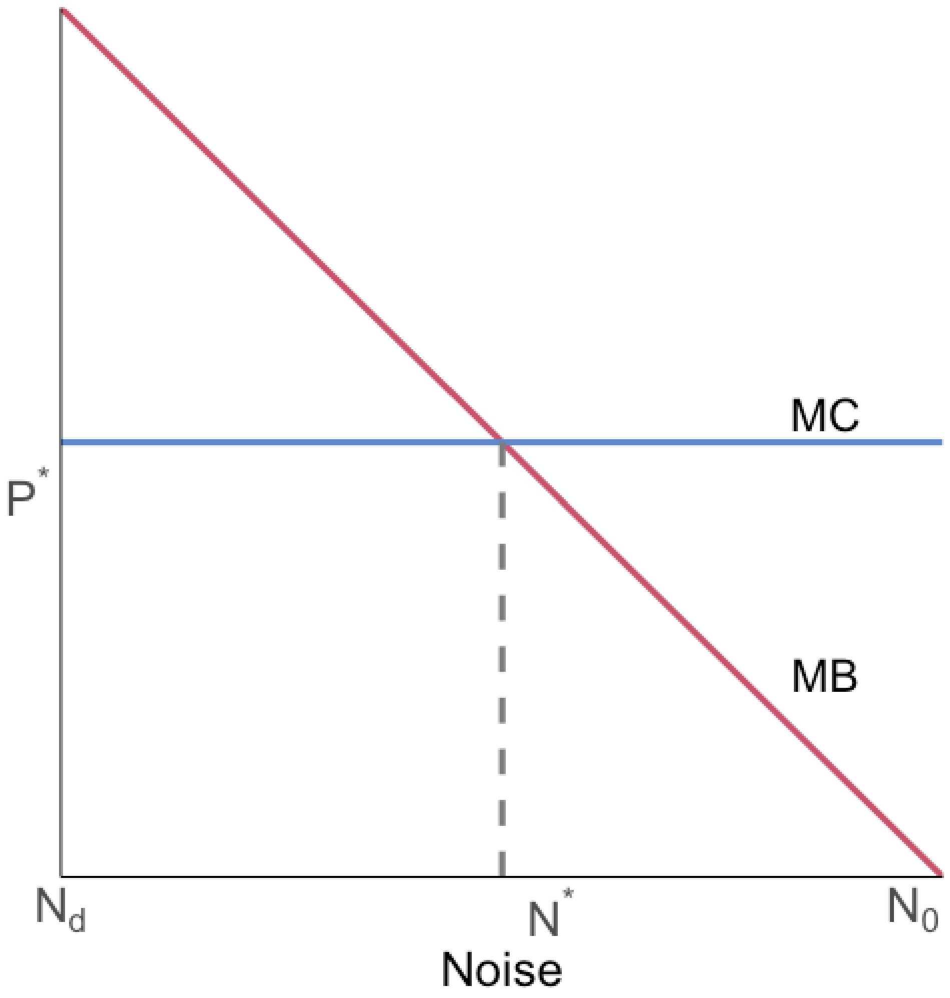


The doctor can pay the data center to reduce operation of its noisy equipment

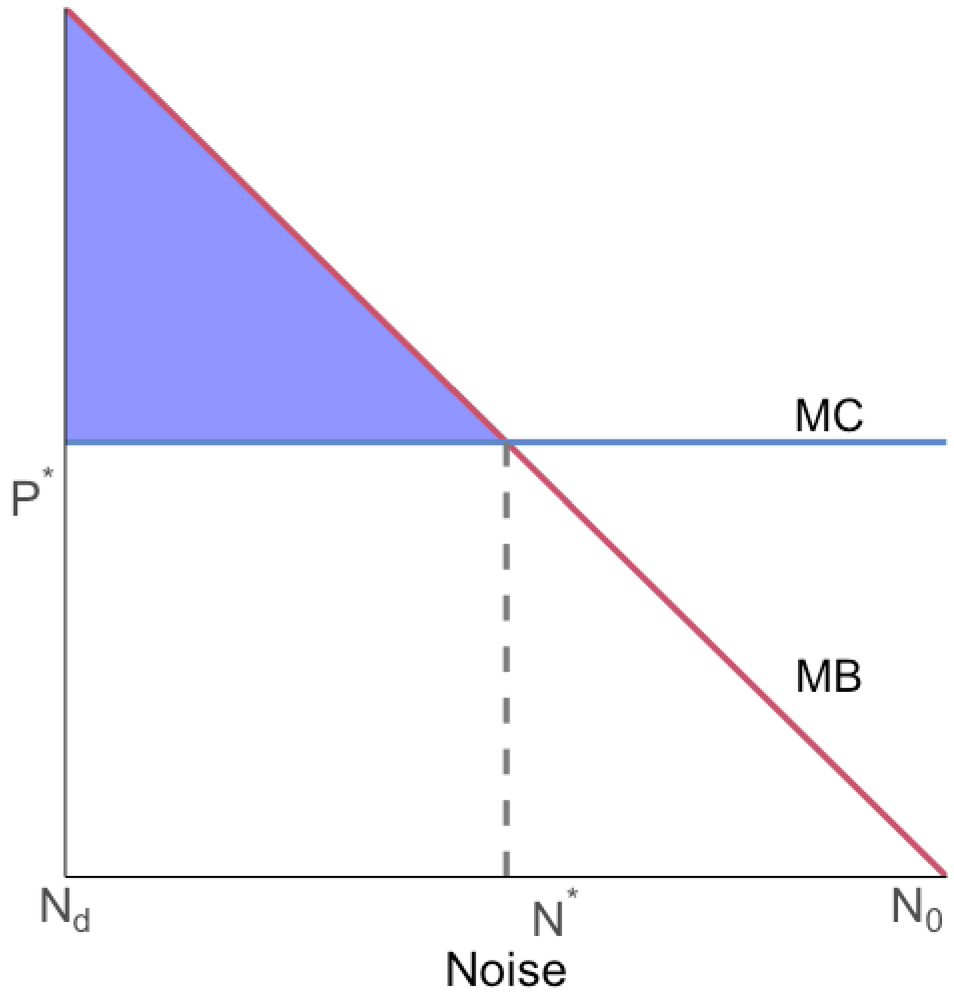
Why?

Because MC to the doctor is higher than MB to the data center for noise at N_0

Coase: Point 1



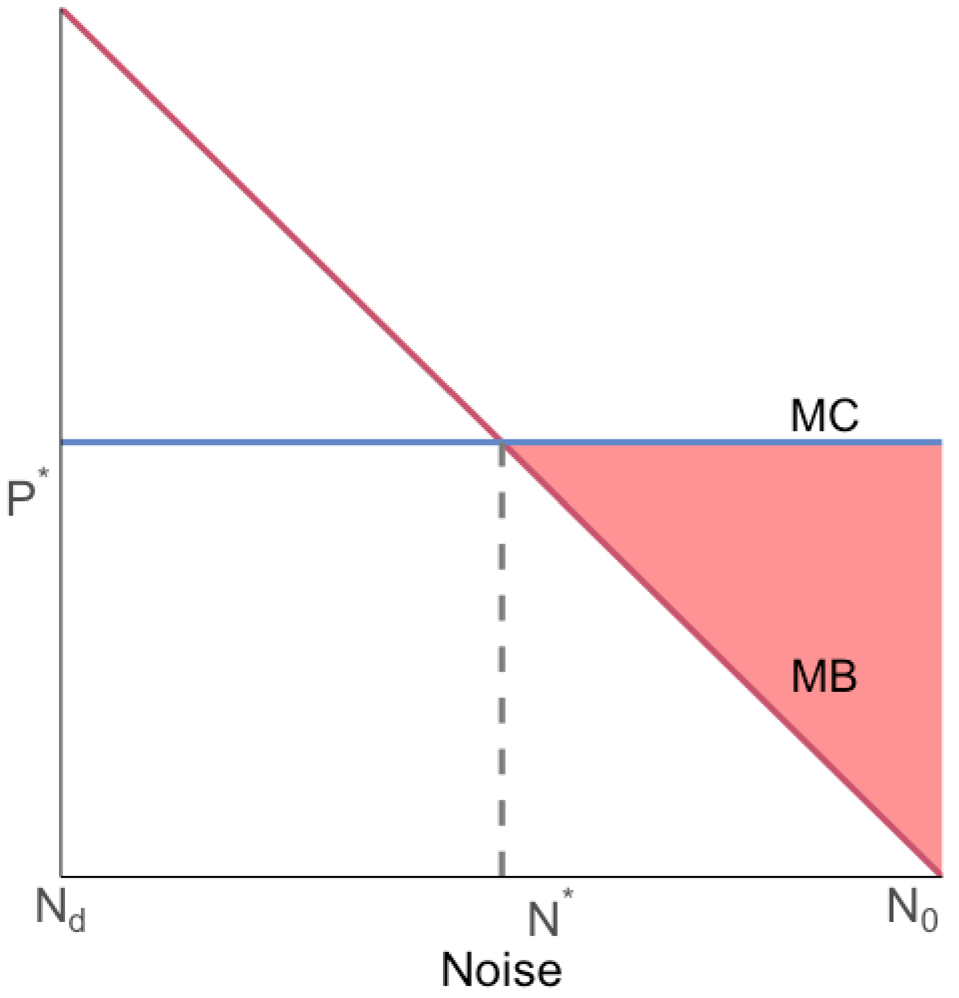
Coase: Point 1



The doctor is willing to pay more (MC) than the data center is willing to accept (MB) until noise is reduced to N^*

This is where total benefit is maximized (blue area)

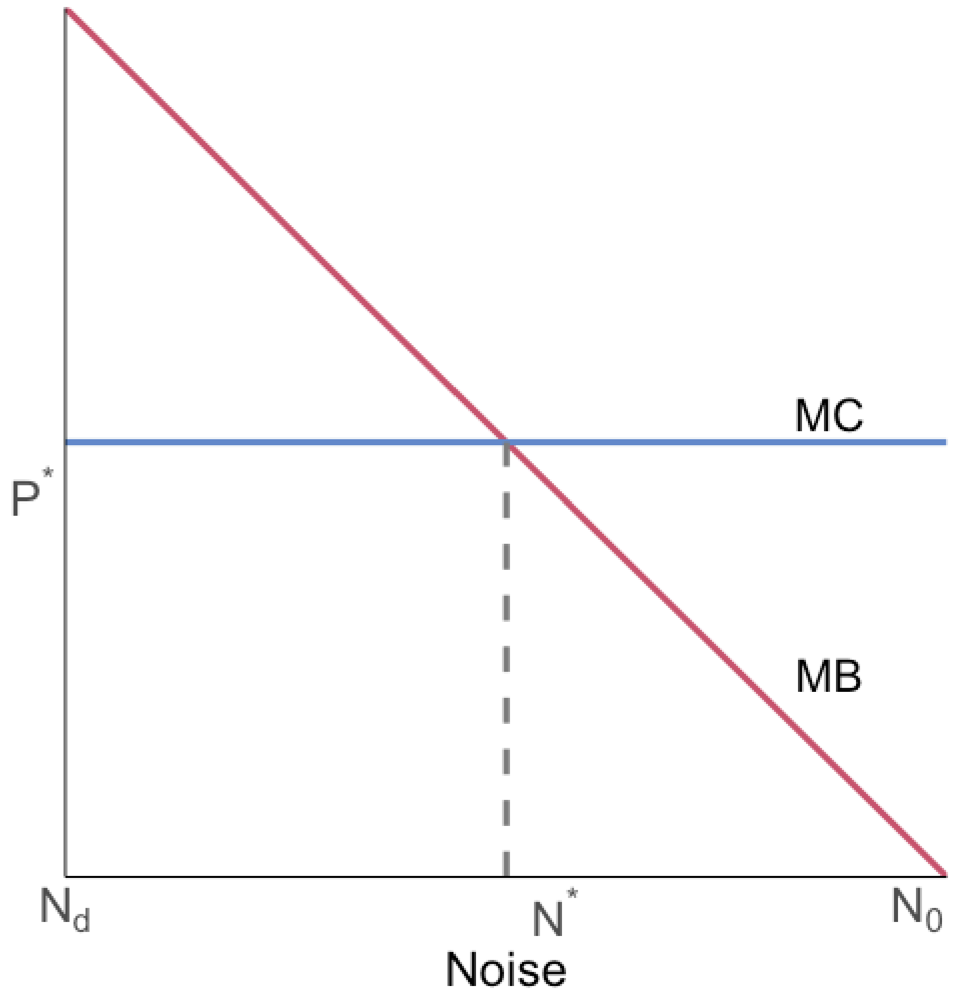
Coase: Point 1



The doctor and data center can split the **bargaining surplus**, the red area

This is just the avoided deadweight loss from the noise externality

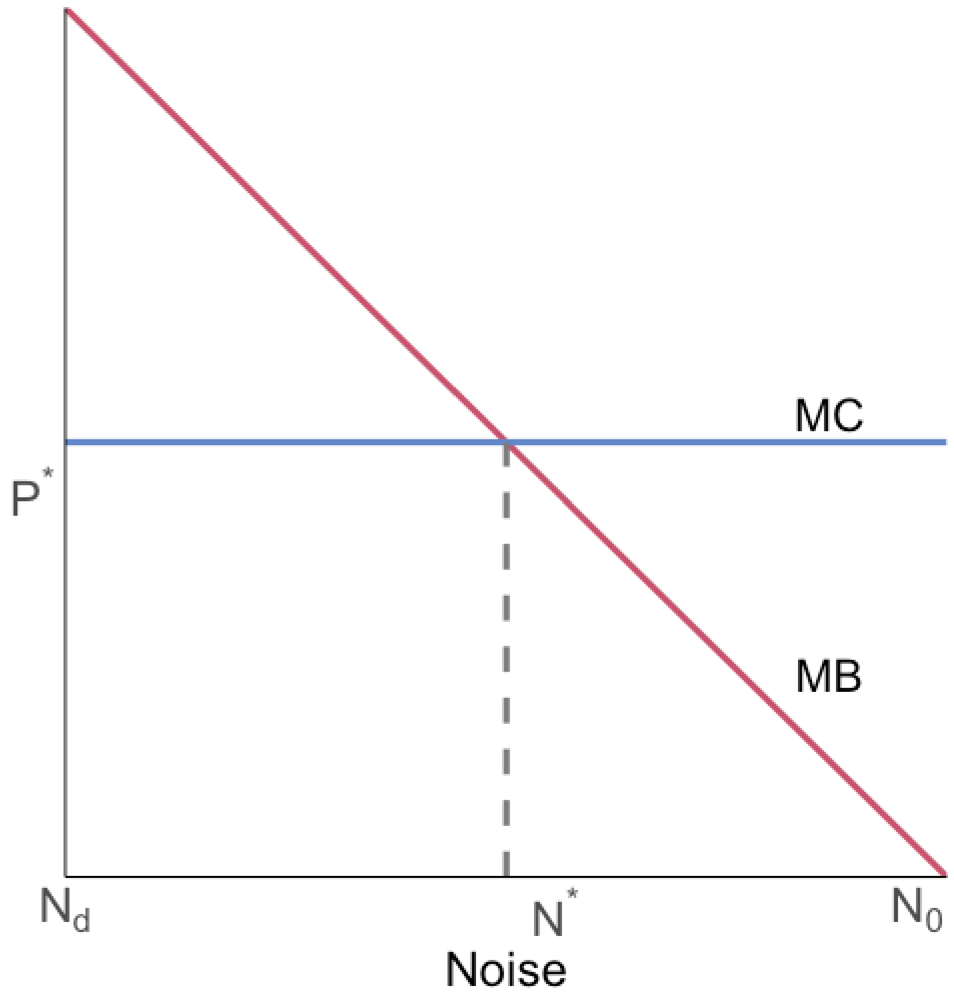
Coase: Point 1



Instead of assigning the right to create noise to the data center, we could give the doctor the right to quiet

In this case what happens?

Coase: Point 1

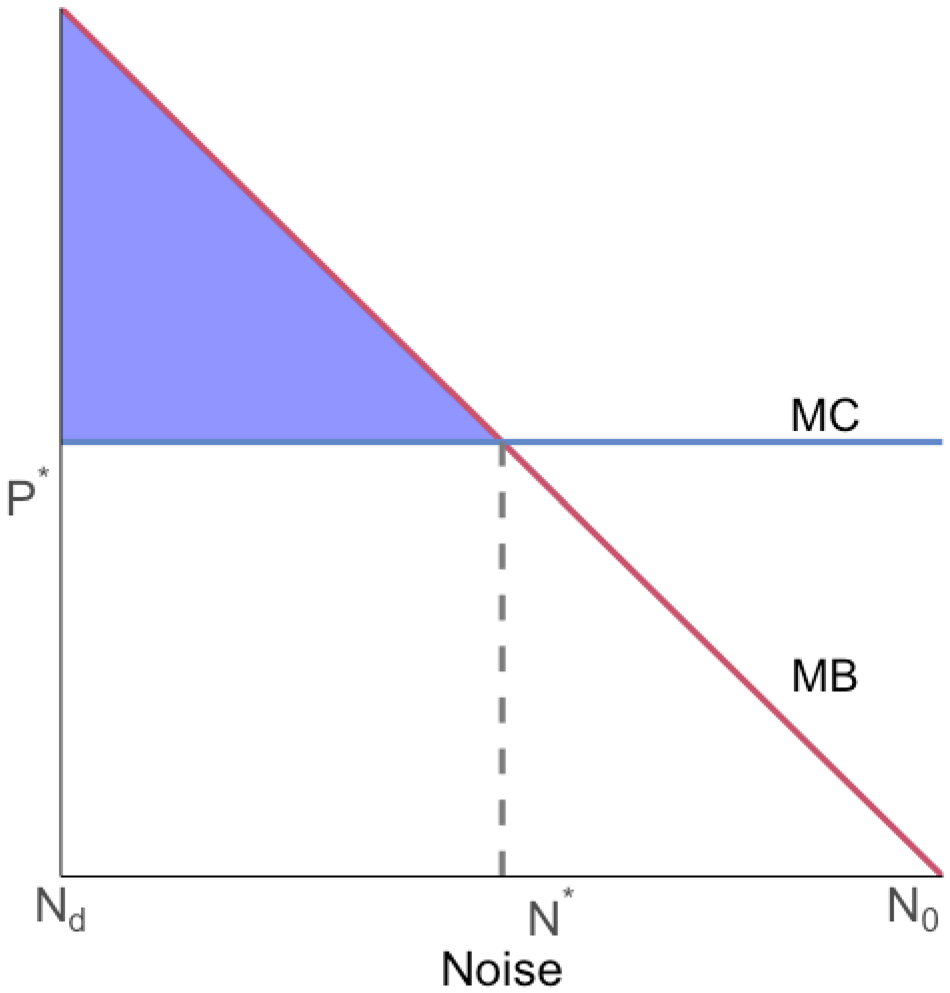


First, we start at N_d now since the doctor does not like noise

The data center pays the doctor to be allowed to make noise

The data center is willing to pay (MB) more than the doctor is willing to accept (MC) until we reach N^*

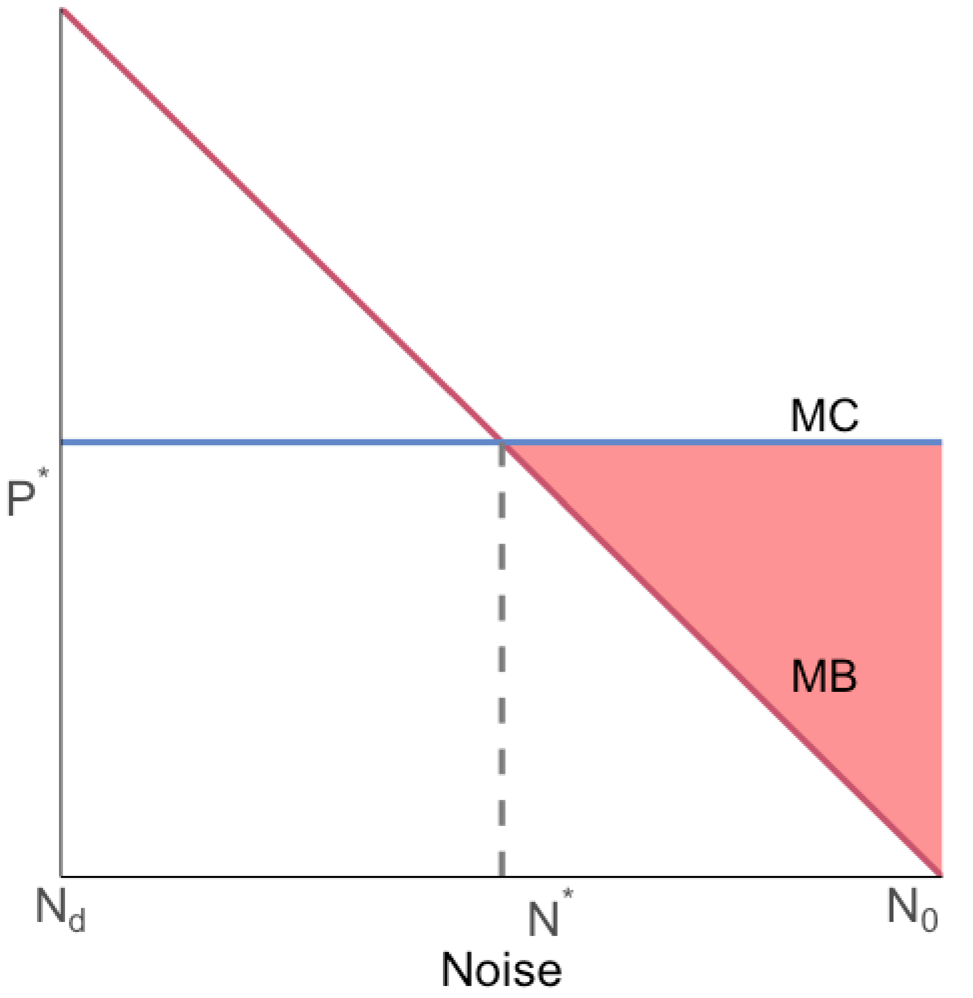
Coase: Point 1



We now maximize surplus (**blue**) and gain bargaining surplus (**blue**) that is split between the doctor and data center

It didn't matter who had the property rights, we managed to get to N^*

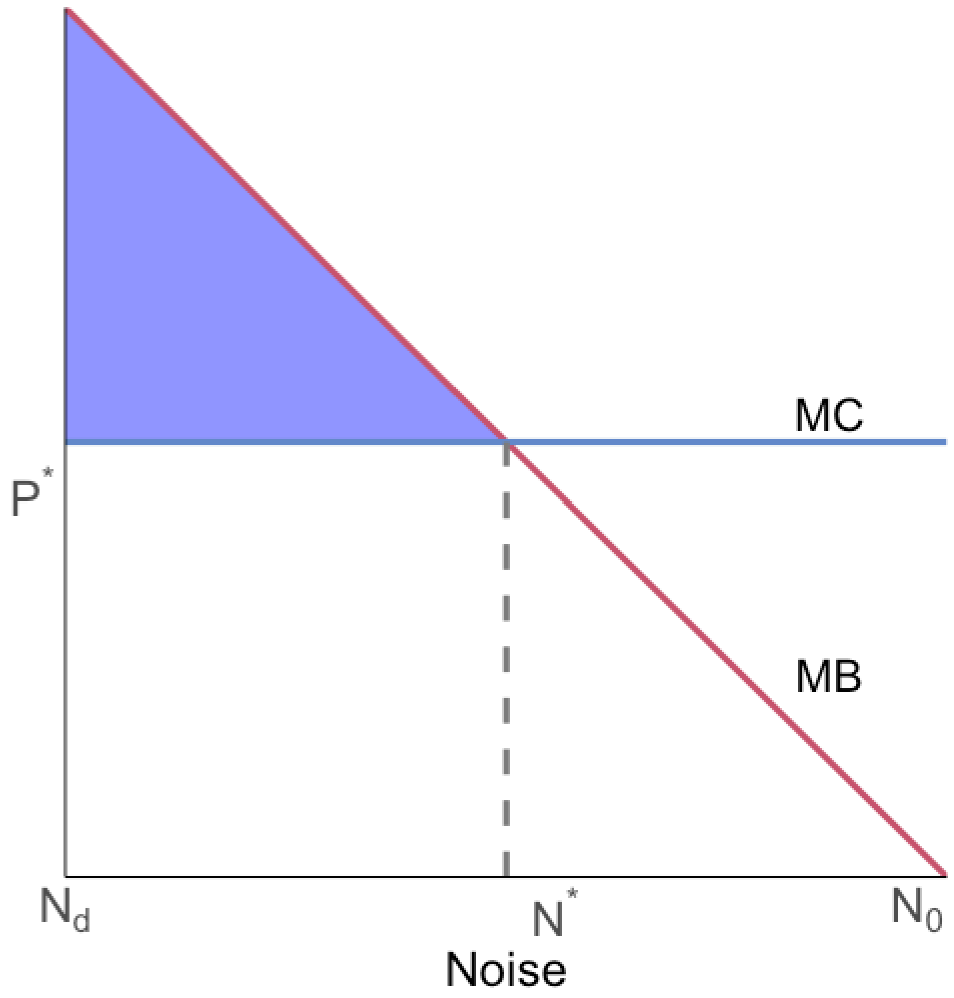
Coase: Point 2



The initial assignment of property rights does matter for the distribution of surplus

If the data center has the right to create noise, it receives a payment from the doctor up to the total size of the red area (bargaining surplus)

Coase: Point 2



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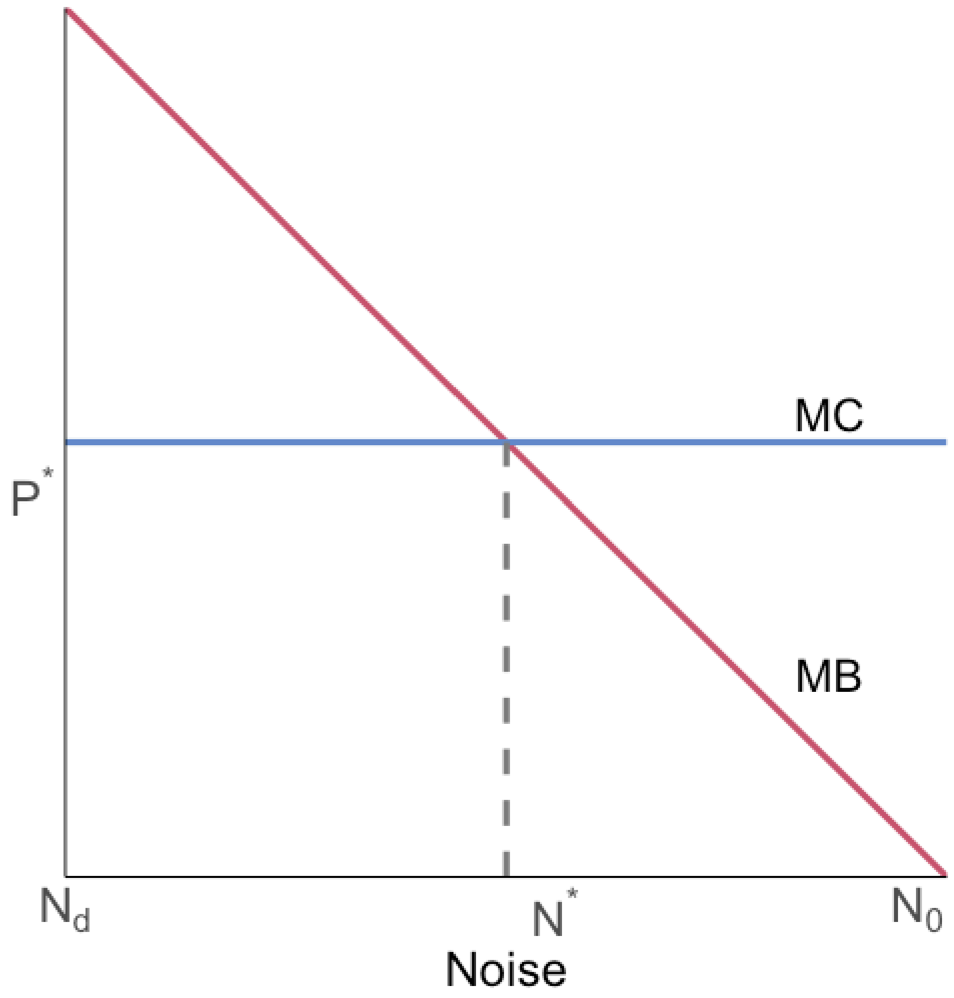

This means that property rights are valuable!

If you have property rights, others have to incentivize you in order to deviate from your privately optimal choice

You will only change the level of noise if your welfare/surplus improves

Coase Caveats

Coasean bargaining does not always work like you think

There are two key pieces we need to have satisfied:

1. No Transactions costs
2. No wealth/income effects

Coase Caveats: Transactions costs

Suppose the doctor owns the right to quiet

Noise N imposes cost $C(N)$ on the doctor and produces benefit $B(N)$ for the data center

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Status quo is no contract: no additional noise and no transfer payment

Coase Caveats: Transactions costs

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Status quo is no contract: no additional noise and no transfer payment

This is a one-shot game: the data center makes one take-it-or-leave-it offer in which the doctor accepts noise N in exchange for payment P

Coase Caveats: Transactions costs

Timing:

1. The data center pays transaction/setup cost tr to propose one contract (N, P)
2. Doctor accepts or rejects once
3. If rejected, they stay at the status quo (no additional noise, no transfer)

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Timing:

1. The data center pays transaction/setup cost tr to propose one contract (N, P)
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There is no renegotiation after the accept/reject decision

Coase Caveats: Transactions costs

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What contract does the data center offer in equilibrium?

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What contract does the data center offer in equilibrium?

i.e. what contract proposal maximizes the data center's profit?

Coase Caveats: Transactions costs

The data center will choose to offer $P = C(N)$

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The realized payment in that interval depends on bargaining power and bargaining protocol

Coase Caveats: Transactions costs

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$$\pi(N) = B(N) - \underbrace{C(N)}_{P} - tr$$

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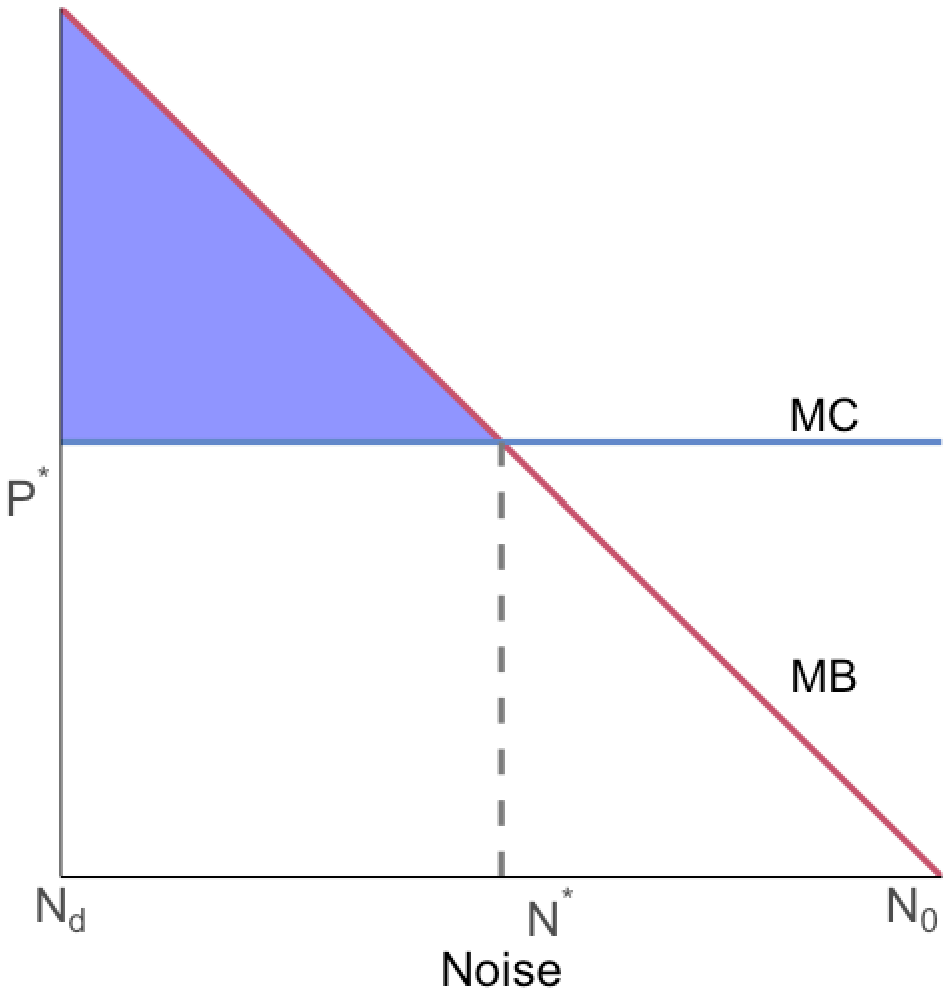
Did transactions costs actually cause any problems?

Coase Caveats: Transactions costs



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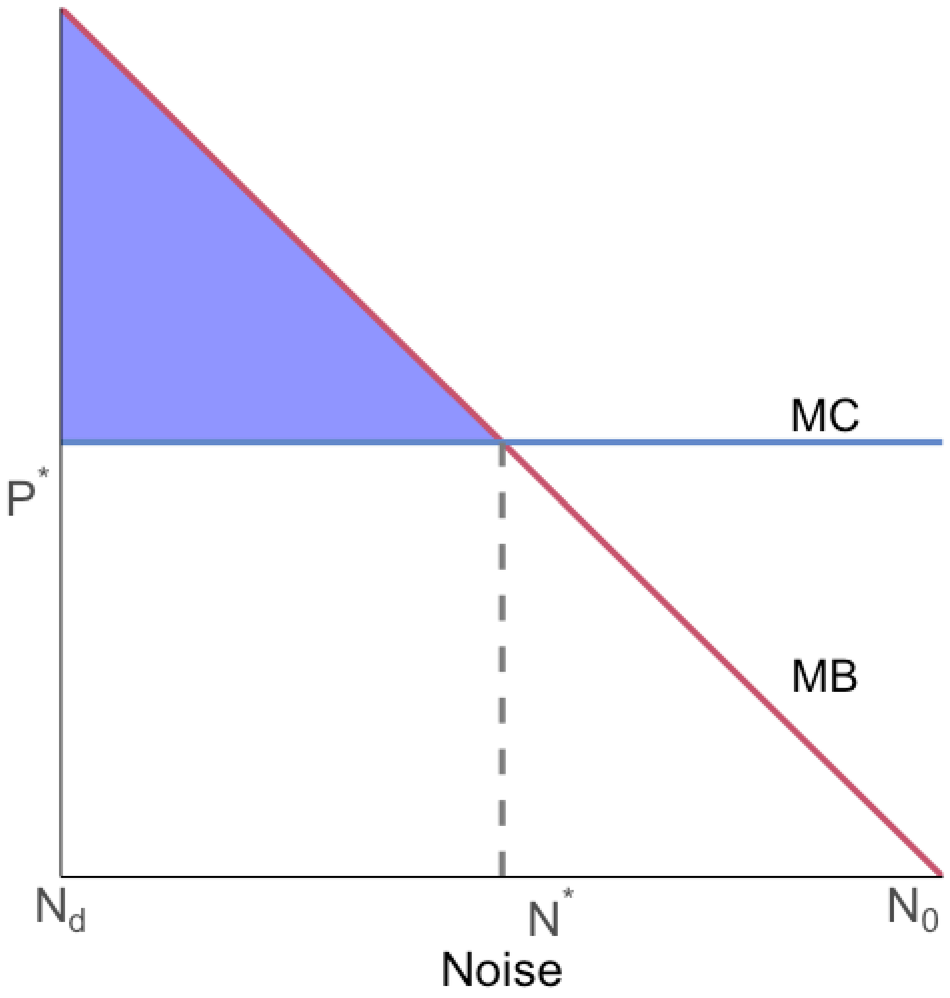
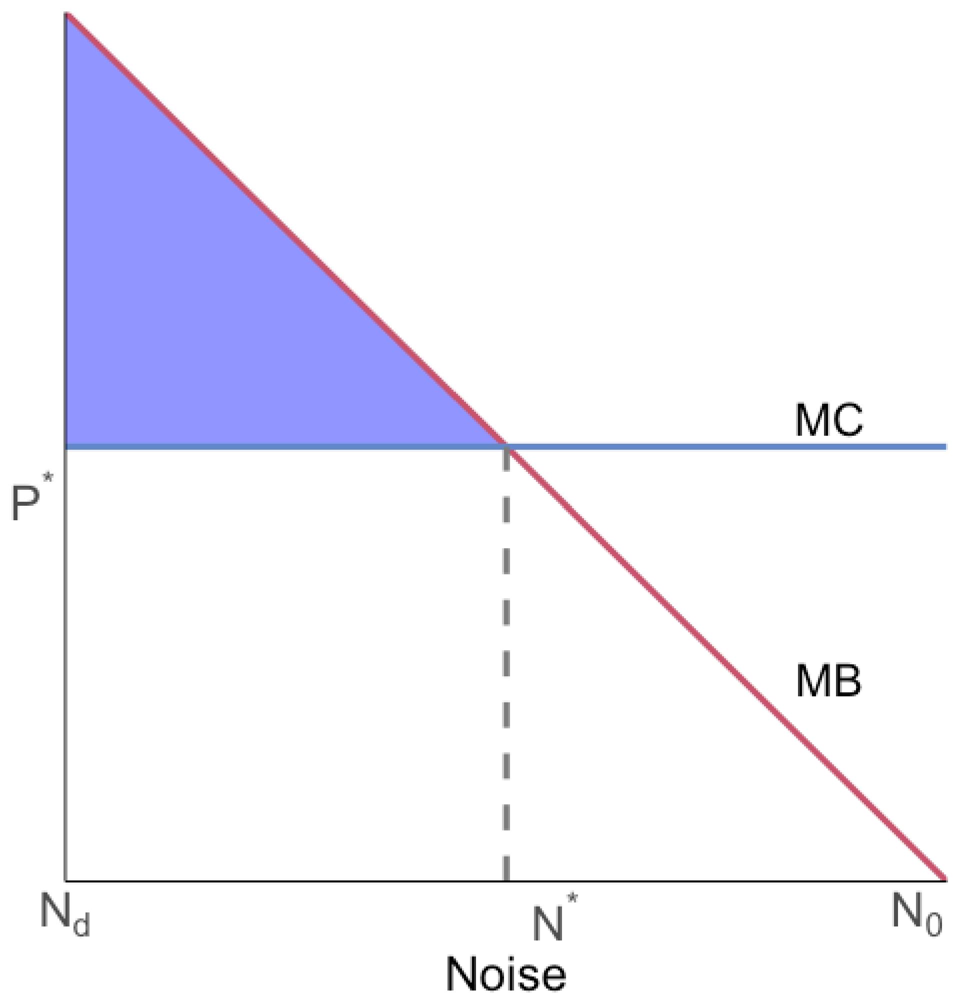
Coase Caveats: Transactions costs



Yes! Why?

To have a mutually beneficial contract we still need the total gain in surplus (**blue**) to be greater than tr

Coase Caveats: Transactions costs



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Otherwise the total cost of the bargaining is greater than the total benefit from bargaining $\rightarrow$ bargaining makes us worse off

Coase Caveats: Transactions costs

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The doctor makes a one-shot take-it-or-leave-it contract offer (N, P) in which the data center reduces noise from N_0 in exchange for a payment

When does the data center accept?

The data center accepts or rejects once; if rejected, noise stays at N_0 with no transfer

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When does the data center accept the contract?

The data center accepts if $P \geq B(N_0) - B(N)$ (payment exceeds its lost operating benefit)

Under one-shot take-it-or-leave-it bargaining, the doctor offers the minimum required: $P = B(N_0) - B(N)$

The feasible payment range

With unlimited bargaining, for a given N, the feasible payment range is between the minimum the data center is willing to accept and the maximum the doctor is willing to pay

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Again, the realized payment in that interval depends on bargaining power and bargaining protocol

Coase Caveats: Transactions costs

The doctor chooses N to minimize total cost:

$$\min_N C(N) + \underbrace{[B(N_0) - B(N)]}_{P} + tr$$

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Coase Caveats: Transactions costs

The doctor chooses N to minimize total cost:

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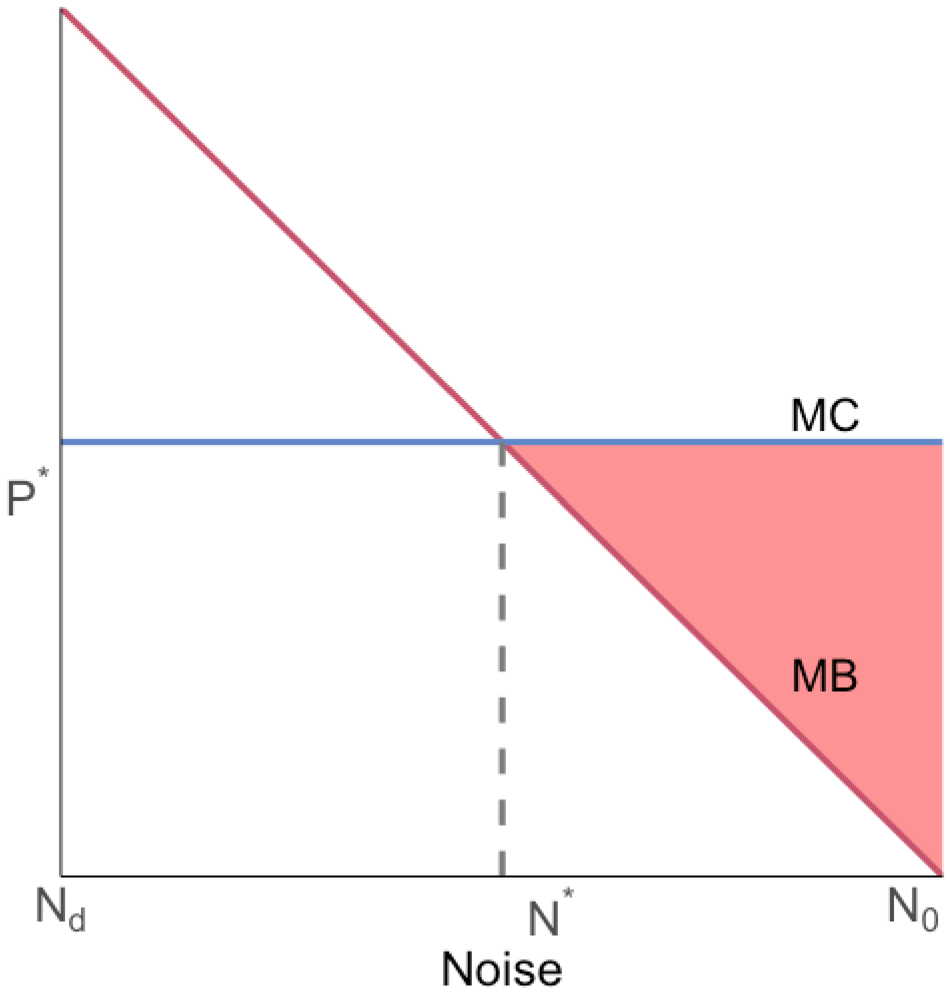
$C(N)$ is the remaining noise cost, P is the payment, and tr is the transaction cost

The first-order condition is:

$$C'(N) = B'(N)$$

Coase Caveats: Transactions costs



We again reach the social optimum!

To have a mutually beneficial contract we still need the total gain in surplus (**red**) to be greater than tr

Otherwise the total cost of the bargaining is greater than the total benefit from bargaining $\rightarrow$ bargaining makes us worse off

Coase Caveats: Transactions costs

Main takeaway: transactions costs can prevent Coasean bargaining from achieving the efficient allocation

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Main takeaway: transactions costs can prevent Coasean bargaining from achieving the efficient allocation

Why?

In cases where the gains from bargaining are small, transactions costs may exceed the benefits and prohibit bargaining from occurring

Coase Caveats: Wealth effects

Changing the initial endowment can change our outcome if wealth changes the doctor's preferences for noise

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The property right is valuable: the owner is richer even before trading

Quiet makes the doctor more productive: owning the property right raised his wealth before negotiating

Coase Caveats: Wealth effects

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With diminishing marginal utility of wealth, extra dollar matters less:

- If the doctor owns quiet, he requires a larger payment to accept noise
- If he does not own quiet, he is not willing to pay as much to reduce the noise

Coase Caveats: Wealth effects

Suppose the data center gains \$90 from noisy operation

When the doctor owns the right to quiet he is wealthier and values the noise at \$100, but when the doctor does not own the right to quiet he is poorer and values the noise at \$80

- Doctor owns quiet (wealthier): WTA to accept noise = \$100 $\rightarrow$ no trade $\rightarrow$ quiet
- Data center owns the right to create noise (doctor poorer): WTP for quiet = \$80 $\rightarrow$ no trade $\rightarrow$ noise

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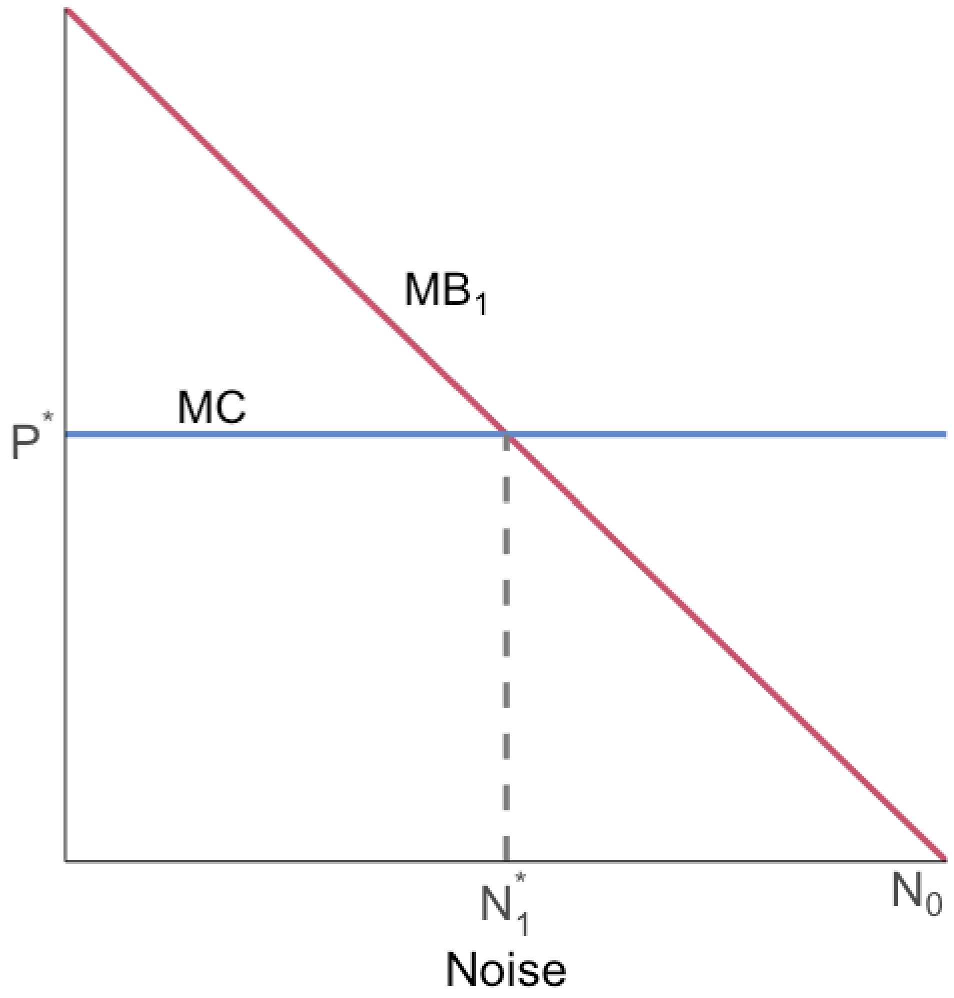
- Doctor owns quiet (wealthier): WTA to accept noise = \$100 $\rightarrow$ no trade $\rightarrow$ quiet
- Data center owns the right to create noise (doctor poorer): WTP for quiet = \$80 $\rightarrow$ no trade $\rightarrow$ noise

Different initial rights $\rightarrow$ different final allocation

Coase Caveats: Income effects



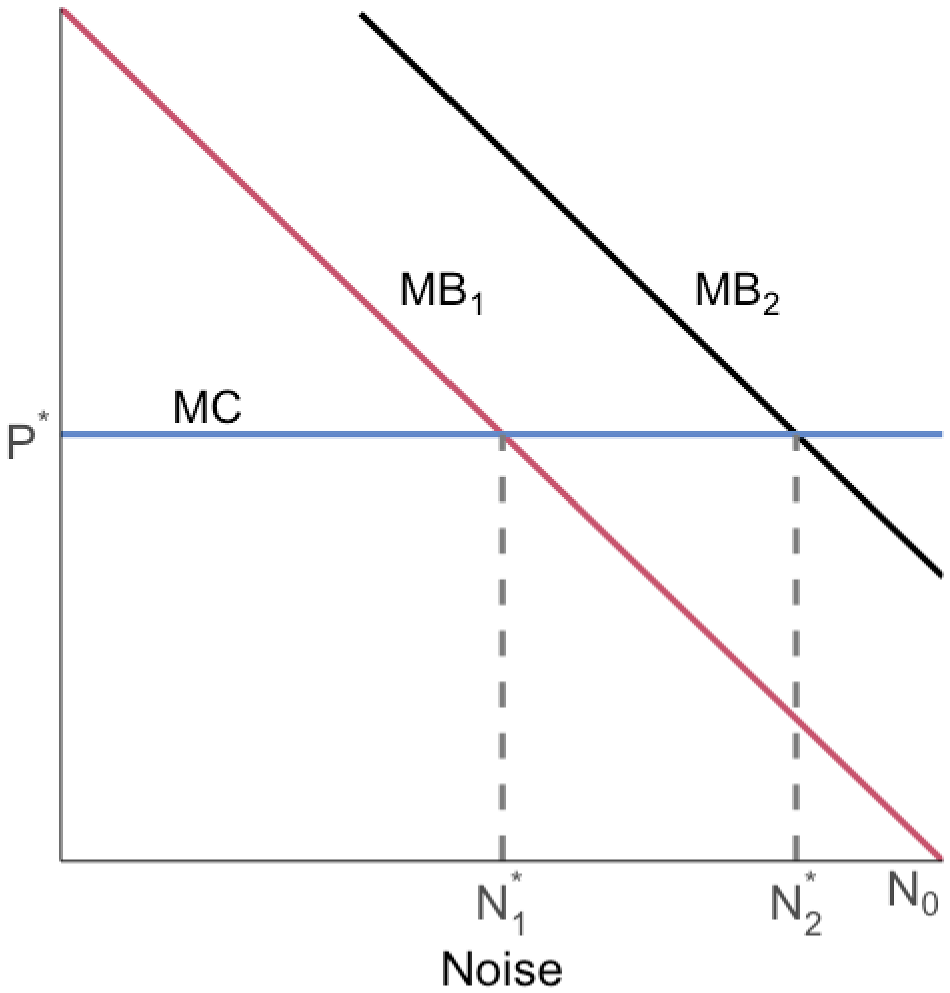
Suppose the data center is given the right to create noise (start at N_0)

The doctor pays the data center to reduce noise and move to N_1^*

The data center receives some bargaining surplus

What can the data center do with it?

Coase Caveats: Income effects



The data center can buy more servers, increasing its marginal benefit from operating the noisy cooling and power equipment

This changes the optimal noise to N_2^*

We have a new equilibrium!

If they contracted to reach N_1^*, it is now **inefficient**

Ways to alleviate transactions cost

One common issue is incomplete information

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Disseminating information can make it easier to know each other's costs and benefits which makes beneficial trades more likely to occur

Ways to alleviate transactions cost

Emergency Planning and Community Right-to-know Act (1986) set-up the Toxic Release Inventory (TRI): http://www.epa.gov/tri/

Green (“eco”) labeling

- Allows companies to learn which firms took positive steps to reduce pollution, and to reward them in the marketplace
- Firms need to be able to increase demand by enough to offset higher costs and raise profits

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Is there a role for the government to be involved in verifying "green" claims? What really constitutes "organic"?

Coase theorem

While transaction costs can be the downfall of Coasian bargaining, there are costs to government policy:

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- Government might set the policy incorrectly

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Coase and Pigou both have trade-offs to their approaches to solving environmental issues

An example: The Cheshire transaction

The Cheshire transaction

Cheshire, Ohio: a small town with a population 221 before 2002

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Enough power for 2 million people (completed 1975)

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Why would this be the case?

The Cheshire transaction

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What happened next? Some real world Coasean bargaining

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April 16, 2002: AEP announced its plan to acquire the incorporated town for \$20 million

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Others wanted the plant to compensate them for diminished property value and to address health concerns

April 16, 2002: AEP announced its plan to acquire the incorporated town for $20 million

September 24, 2002: AEP announces that it has finalized the buyout. About 90 percent of town residents have participated in the buyout offer and have signed the health waivers and the confidentiality agreements

The Cheshire transaction

Property owners in town receive 3.5x assessed value

Outside town: 2x assessed value

Renters receive $5k for each year lived in Cheshire, up to $25k

Must sign a health waiver prohibiting them from suing AEP for future health problems

The Cheshire transaction

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Attorneys take about 1/3 of settlement money

The Cheshire transaction: was it a good thing?

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Now imagine that AEP Gavin's control costs were low, and the efficient outcome would be to install additional control equipment

Would it matter whether we granted the "right to clean air" to the town or to AEP?

Data centers: Coasean bargaining in the AI boom

Cheshire, twenty years later

The Cheshire transaction was one power plant buying out one small town

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The difference is exactly what our transactions cost model predicts

A bargaining attempt: Ashburn, Virginia

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But the HOA president told NBC4 there was **no current offer**, and no sale closed

Source: NBC4 Washington, March 25, 2026.

Why didn't a deal close?

- **Many parties:** 143 homeowners must coordinate
- **Holdouts:** each owner may demand a larger share of the surplus
- **Regulatory risk:** purchase does not guarantee data center approval

When the deal doesn't close: holdouts

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But bargaining with many owners is fragile:

- Is a "no" true WTA, or a holdout for a bigger share of the surplus?
- With N owners who must all say yes, transactions costs rise fast

Why bargaining fails: Southaven

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Why is there no Cheshire-style transaction here?

- **Many parties:** thousands of households breathe the emissions
- **Contested rights:** who owns the air? Permits are disputed, not settled
- **Free riding:** every resident hopes the others fund the fight
- **Information:** health damages are diffuse and hard to document

What replaces bargaining? Courts and regulators

High transactions costs choke off bargaining $\rightarrow$ the conflict moves to courts and regulators

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The lawsuit concerns 27 allegedly unpermitted turbines at the Southaven power plant

Instead of bargaining over emissions, the parties are asking courts and regulators to define and enforce the rights

Data centers through Coase's lens

	Ashburn proposal	Farmland holdout	Southaven gas plant
Parties	Few	Few	Thousands
Property rights	Clear (land)	Clear (land)	Contested (air)
Transactions costs	Low	Medium	High
Outcome	No completed deal	No deal	Federal lawsuit

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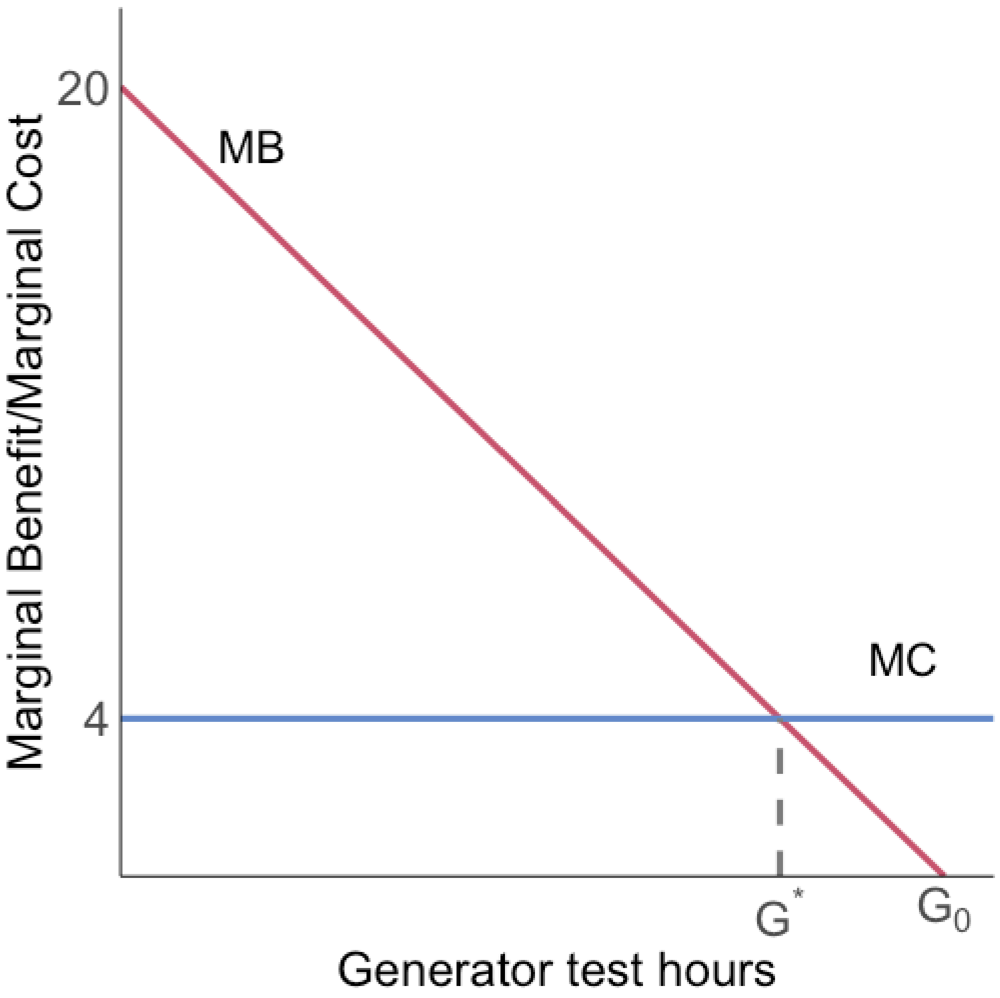
Same technology, same externalities, three different outcomes

The Coase theorem tells us where to look: **who holds the rights, and how costly is it to strike the deal?**

Your turn: a data center next door



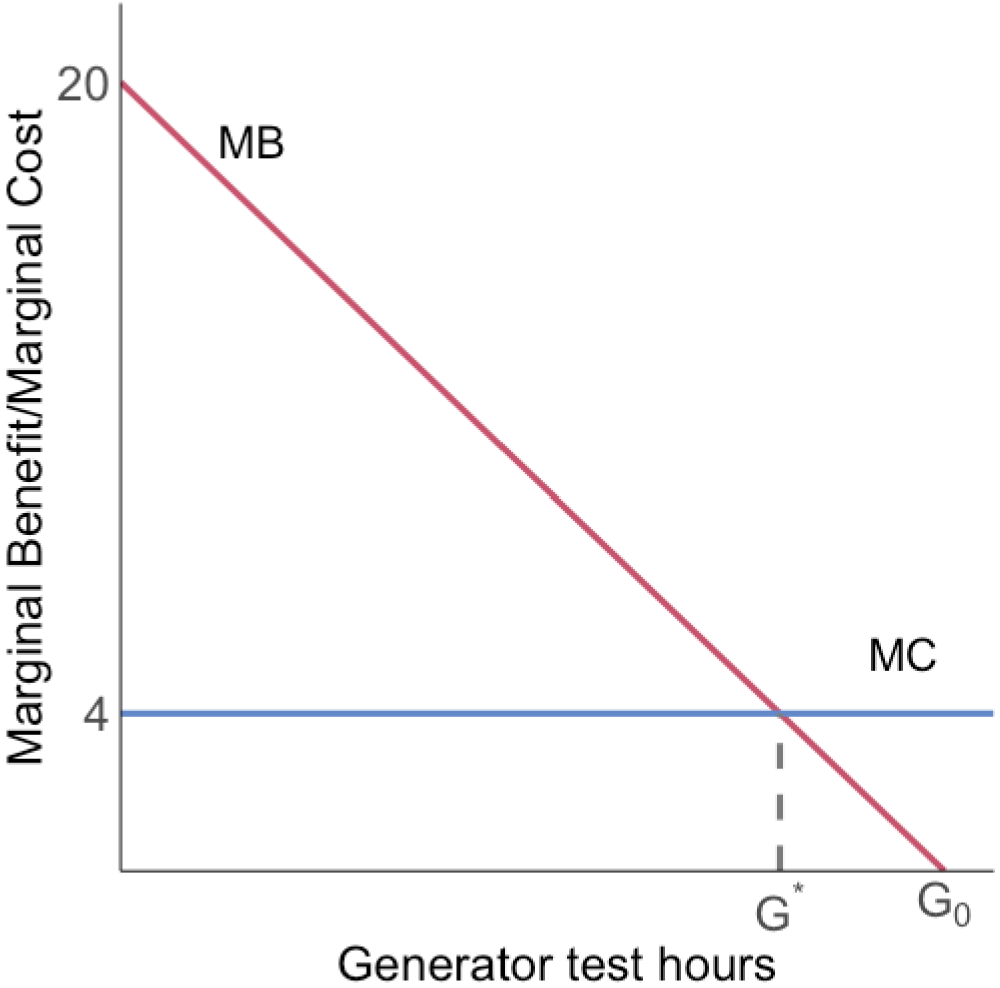
A data center tests its backup generators G hours per month

Marginal benefit to the operator: MB = 20 - 2G

Marginal noise cost to the residents: MC = 4

Left alone, the operator would run $G_0 = 10$ hours; residents want zero

Your turn: a data center next door



MB = 20 - 2G, MC = 4

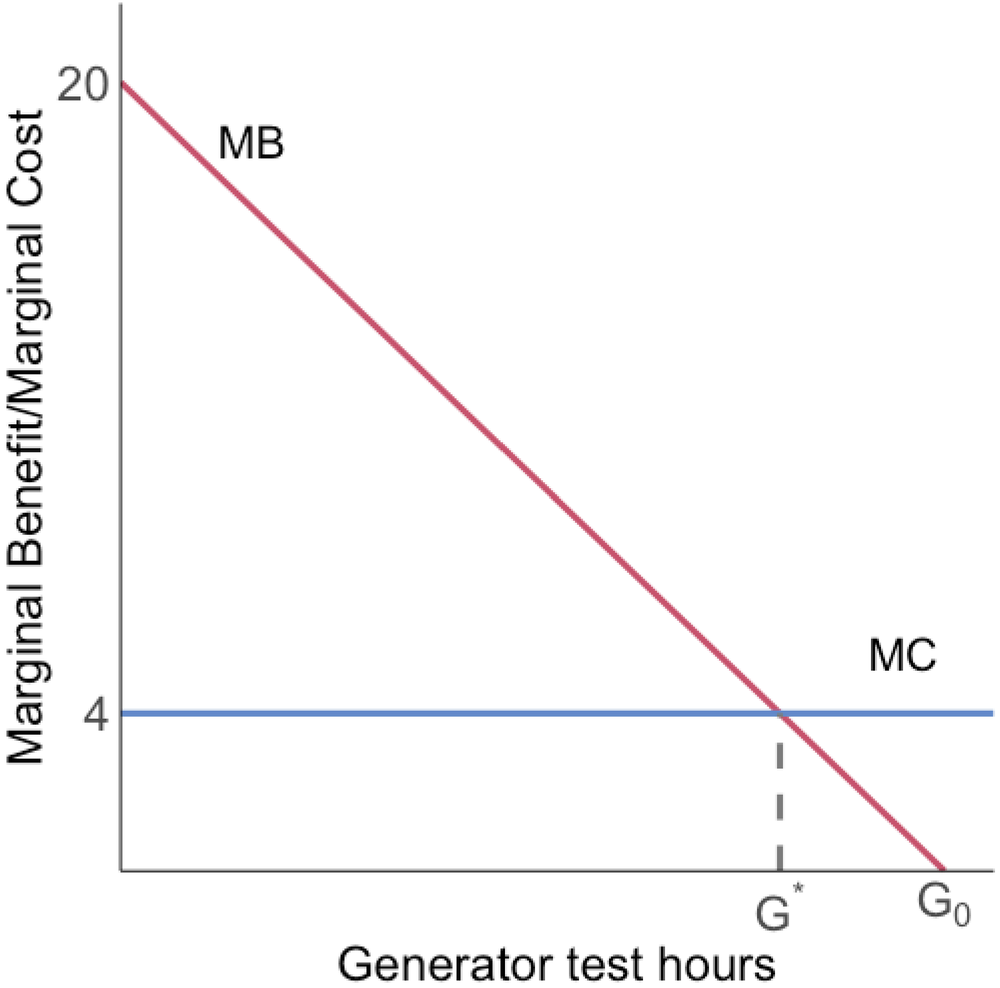
Any deal needs lawyers:

transactions cost $tr = 10$

Your turn:

1. Find the efficient hours G^*
2. Rights to the residents: what payments could work?
3. Under each rights assignment: does a deal happen?

Your turn: the efficient hours

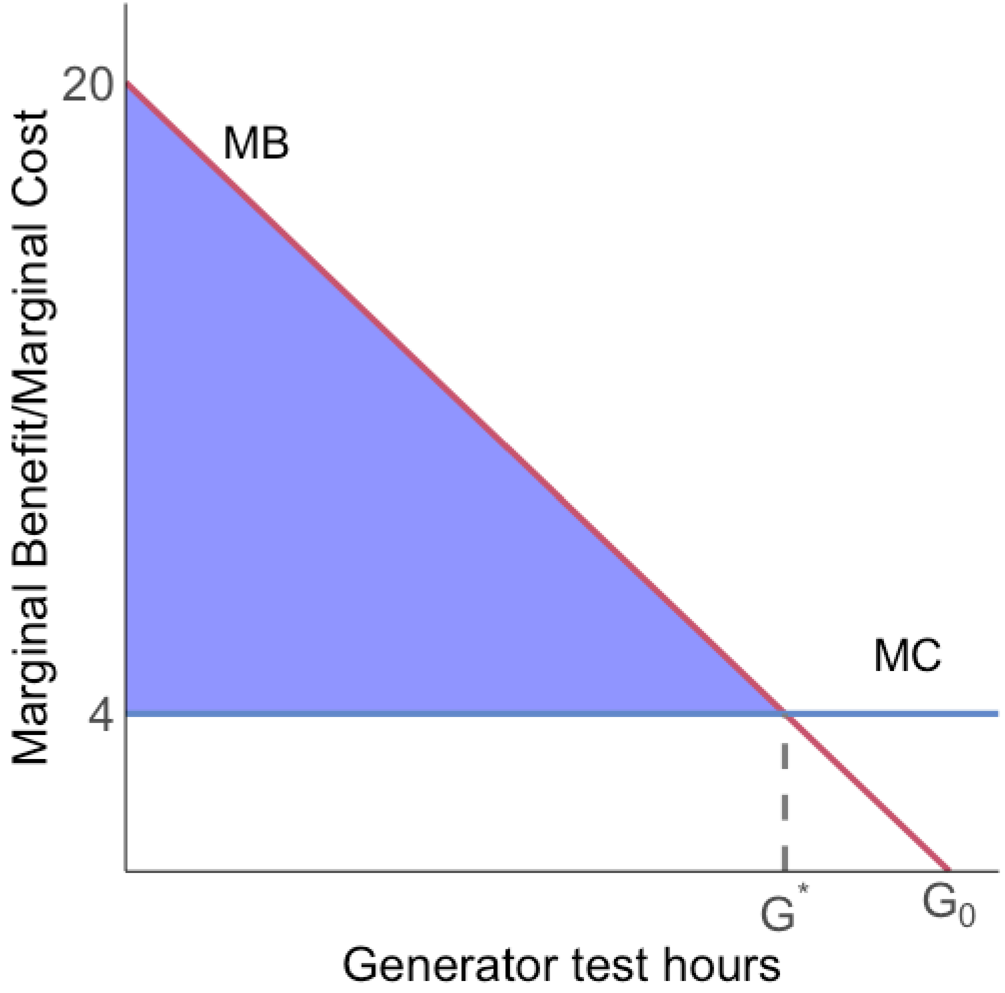


Efficiency: MB = MC

20 - 2G = 4 $\Rightarrow G^* = 8$

The first 8 hours are worth more to the operator than they cost the residents; the last 2 are not

Your turn: residents hold the rights



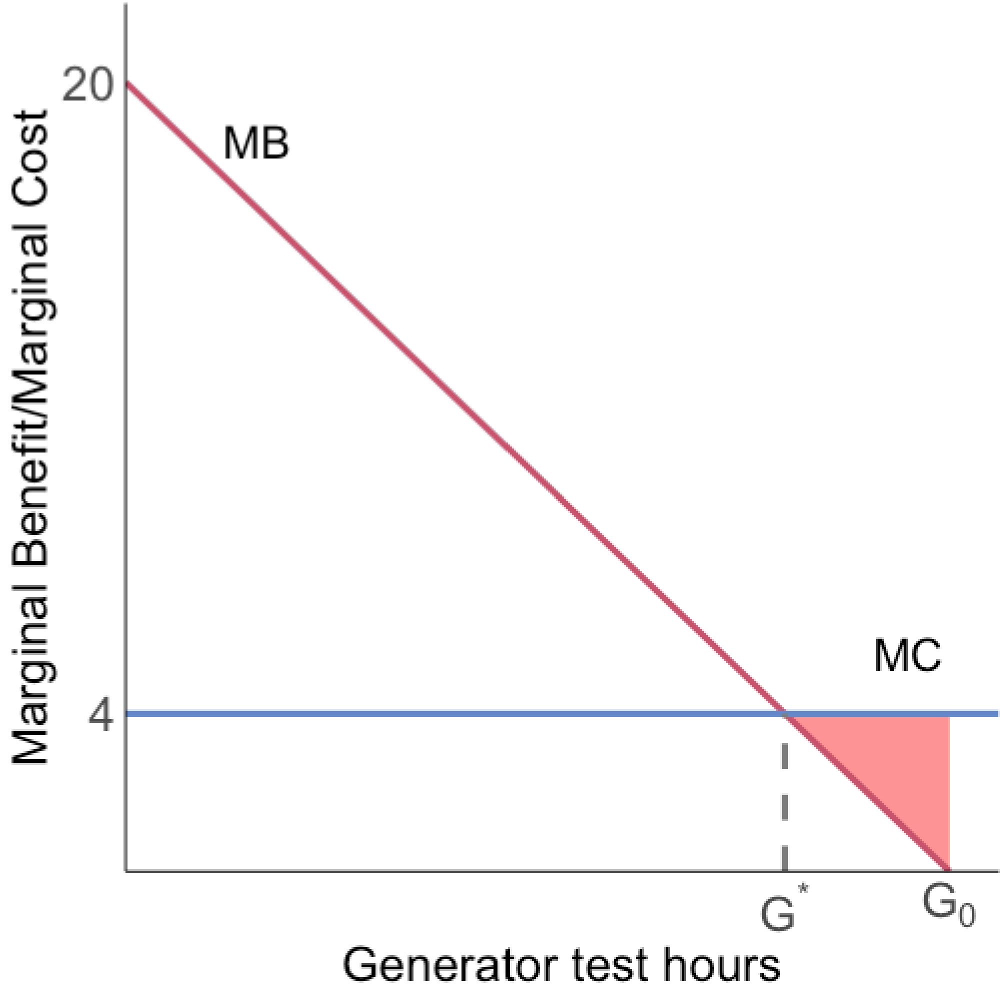
Start at G = 0; the operator pays for hours

Residents accept $P \geq 4 \times 8 = 32$; the operator pays at most his benefit, $P \leq 96$

Gains from trade: the **blue** triangle $= \frac{1}{2} \times 8 \times 16 = 64$

$64 > tr = 10 \rightarrow$ **the deal happens** and we reach $G^* = 8$

Your turn: the operator holds the rights



Start at $G_0 = 10$; residents pay to cut hours

Gains from trade: the **red** triangle $= \frac{1}{2} \times 2 \times 4 = 4$

$4 < tr = 10 \rightarrow$ **no deal**, and skipping the deal is the right call

With transactions costs, **who holds the rights changes the outcome**

Let's bargain: the data center game

The data center bargaining game

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Your payoffs across all three rounds go on the class leaderboard

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We will check the class results against these predictions after we play

Coase in practice

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The cap-and-trade system will then achieve the efficient outcome